

A Finite-Hilbert-Space Substrate for Emergent Constraints and Bell Correlations

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This work investigates how quantum-like correlations and classical-looking structure can arise inside finite-dimensional quantum systems equipped with local Hamiltonians and constraint operators. The starting point is an explicitly finite Hilbert space built from local matter degrees of freedom and auxiliary “gauge” qubits, evolved unitarily by a local Hamiltonian. No assumptions are made about the structure of the Standard Model, spacetime, or quantum gravity. All claims concern only the behaviour of well-specified toy models that can be simulated exactly.

The paper has two goals. First, it formulates a conservative “substrate framework” in which what we call a system is viewed as a dynamically stable pattern in a finite Hilbert space subject to local constraints. Second, within this framework it studies how Bell-type correlations respond to changes in constraint strength and to attempts to extract information from the substrate. Numerical experiments on small lattices show that (i) purely constraint-driven dynamics can generate CHSH violations up to the Tsirelson bound, (ii) these violations are robust across wide parameter ranges, and (iii) both generic perturbations and explicit information extraction degrade the Bell correlations in a way that correlates with entropy production in the substrate.

As a complementary demonstration, a three-quark “proton” toy model is constructed on a finite lattice using the same constraint language. Its ground state exhibits a stable, centrally-peaked density profile and nontrivial spin statistics, illustrating how bound-state-like structure can emerge in the same framework. Throughout, a conservative posture is maintained: if the data do not show it, we do not say it.

I. INTRODUCTION

Modern quantum theory is defined as much by the *constraints* it imposes as by the dynamical laws it supplies. The standard formalism combines a handful of structural requirements—linear Hilbert space, unitary evolution generated by a Hermitian Hamiltonian, and the Born rule—with empirical constraints such as no-signalling and the Tsirelson bound on Bell inequality violations. Taken together, these carve out a narrow region of correlation space: classical hidden-variable models lie below the Bell bounds, while generic nonlocal theories exceed the quantum limit. The empirical success of quantum mechanics suggests that these constraints encode real structure, but it is not obvious why the physical world realizes exactly this pattern and not something more extreme.

The *substrate framework* studied here takes a deliberately modest approach to that question. Instead of postulating new primitives beyond standard quantum theory, it starts from finite-dimensional Hilbert spaces, local Hamiltonians, and explicit constraint operators, and asks what kinds of effective behaviour can emerge. The focus is not on reproducing the Standard Model or gravity, but on understanding how classical-looking patterns and quantum-like correlations can arise inside small, exactly simulable models.

The program is organized into three tiers: (i) non-speculative mathematical definitions, (ii) numerical or structural observations, and (iii) explicitly flagged interpretive hypotheses. Only tier (ii) is used as evidence; tier (iii) is presented as conjecture to guide future work. The guiding principle is simple: if the data do not show it, we do not say it.

Within this framework, Bell tests play a central role. Rather than treating the CHSH inequality as a binary test of locality, we use it as a diagnostic of how constrained substrate dynamics balance correlation, energy, and information flow. Concretely, we study a minimal two-site, two-qubit toy model whose coarse-grained behaviour can be mapped to a standard CHSH scenario. The underlying degrees of freedom live on a finite lattice with a defragmentation potential that stabilizes localized patterns and a Gauss-like constraint that energetically suppresses certain occupancy configurations. By sweeping the strength of these constraints and by introducing controlled perturbations and readout channels, we explore how Bell correlations depend on the substrate-level rules.

The central operational question is: *Can a purely constraint-driven substrate, with no built-in Hilbert space structure beyond finiteness and unitarity, generate Bell correlations that violate the classical CHSH bound, saturate the Tsirelson limit, and degrade in a controlled way under information extraction?* The simulations presented below answer this in the affirmative for a specific family of finite models.

A second, complementary question is whether the same substrate language can support bound-state-like structure reminiscent of particles. To probe this, we construct a three-quark “proton” model on a finite lattice using similar constraint ingredients. Its ground state shows a stable central density lump and nontrivial spin statistics, suggesting that the framework is rich enough to host both Bell-type nonclassical correlations and spatially localized composite objects.

The remainder of the paper is organized as follows. Section II summarizes the substrate framework and the finite-Hilbert-space construction. Section III describes the specific toy models and the four Bell-related experiments: a Gauss sweep, an exchange sweep, a sequential perturbation test, and an information-extraction test. Section IV presents the numerical results, including CHSH parameters and correlations between entropy, overlap, and extracted information. Section V outlines the three-quark bound-state example. Section VI formulates an experimentally testable prediction. Section VII discusses implications, limitations, and possible directions for future work.

II. FRAMEWORK

A. Axioms and scope

The substrate program is defined by three conservative axioms.

A1. Finite quantum substrate. The fundamental degrees of freedom live in a finite-dimensional Hilbert space $\mathcal{H}$ built from local factors. There is no continuum limit assumed at the outset and no additional ontic primitives beyond the quantum state.

A2. Unitary evolution. Dynamics are generated by a time-independent Hermitian Hamiltonian H that acts locally on the Hilbert space. The evolution of any closed system is given by the Schrödinger equation with $U(t) = e^{-iHt}$.

A3. Emergent structure. Classical-looking variables, effective fields, or “objects”—including anything that might be called a particle or a pointer—are interpreted as coarse-grained, dynamically stable patterns in the underlying quantum state. There is no additional notion of collapse or classicality beyond this emergent description.

These axioms are meant to stay strictly within orthodox quantum mechanics. The question is how far one can go with only these ingredients, plus specific choices of Hamiltonian and constraint operators, in reproducing features normally associated with quantum theory and classical emergence.

B. Finite-Hilbert substrates

The concrete models used here live on finite lattices. For the Bell-type experiments, a convenient starting point is

$$\mathcal{H} = \mathcal{H}_{\text{sites}} \otimes \mathcal{H}_{\text{links}} \otimes \cdots, \quad (1)$$

where the site factors describe local matter degrees of freedom and the link factors encode $\mathbb{Z}_2$ -like gauge or auxiliary variables. In the simplest cases studied numerically,

- each matter site is a few-level system (for example, a four-dimensional “spinor” space carrying occupation and spin),
- each link carries a qubit interpreted as a discrete gauge field or coherence-control register,
- the total dimensionality is finite, so that small lattices are exactly simulable.

All structure is encoded in the Hamiltonian, which is a sum of terms that act nontrivially only on small regions of the lattice.

C. Constraints and locality

Two structural features have been especially important in the numerical explorations so far: local gauge-like constraints and locality of information routing.

1. Local Gauss-like constraints

At each lattice site s , we define a local operator

$$G_s = (-1)^{n_s} \prod_{\ell \in \text{star}(s)} \sigma^x(\ell), \quad (2)$$

where n_s is the occupation number on site s , the product runs over links incident on s , and $\sigma^x(\ell)$ is a Pauli operator on link ℓ . This structure is directly inspired by $\mathbb{Z}_2$ Gauss laws in lattice gauge theory. States that satisfy $G_s = +1$ at all sites can be interpreted as obeying a local consistency rule linking matter occupancy to adjacent links.

Violation of the constraint is penalized energetically by adding a term

$$H_{\text{Gauss}} = \lambda_G \sum_s (\mathbb{I} - G_s)^2, \quad (3)$$

where $\lambda_G \geq 0$ controls the strength of the penalty. For large λ_G , configurations with inconsistent matter-link patterns become very costly; for $\lambda_G = 0$, the constraint is inactive.

2. Locality and information routing

Because G_s couples matter occupancy to nearby links, it can be viewed as a rule governing how information about local configuration is routed through neighboring degrees of freedom. This interpretation is used purely as a structural lens on the toy models; no claim is made that it reflects a fundamental law of nature. It does, however, motivate the use of Gauss-violation measures as probes of how “crowded” or “overlapping” patterns are in the lattice.

D. Toy models and observables

The Bell-related experiments are carried out on 2×2 spatial lattices with matter sites and $\mathbb{Z}_2$ gauge links. Three classes of initial patterns are considered:

- a single localized skyrmion-like pattern,
- two overlapping patterns placed in the same region,

- two separated patterns placed in different regions.

For each configuration we measure energy expectation values $\langle H \rangle$, total occupation, local and global Gauss-violation measures, and a coarse-grained CHSH parameter S constructed from appropriate spin correlators on opposite sides of the lattice.

Across parameter sweeps, several empirical features appear repeatedly: overlapping patterns tend to exhibit larger total Gauss-violation than separated patterns; increasing λ_G amplifies the energy difference between overlapping and separated states; and for sufficiently large λ_G overlapping states become heavily suppressed relative to separated ones. These are strictly numerical outputs of the finite models; they motivate the Bell-focused experiments but are not taken to imply anything about real fermionic statistics or the Pauli principle.

III. METHODS

This section summarizes the concrete numerical experiments performed within the substrate framework. All simulations use finite Hilbert spaces with explicit Hamiltonians and constraints. Where possible, parameters are chosen so that the models are small enough to permit exact diagonalization.

A. Bell experiment mapping

To connect the substrate models to a CHSH scenario, we identify two spatially separated “measurement regions” A and B on the lattice. Each region supports two effective measurement settings, represented by different local operators A_0, A_1 on region A and B_0, B_1 on region B . A given run of the toy model corresponds to preparing an initial state $|\psi_0\rangle$, evolving under the substrate Hamiltonian for a fixed time T , and then evaluating spin correlators

$$E_{ij} = \langle \psi(T) | A_i B_j | \psi(T) \rangle, \quad i, j \in \{0, 1\}. \quad (4)$$

The CHSH parameter is then

$$S = E_{00} + E_{01} + E_{10} - E_{11}. \quad (5)$$

Classical local-hidden-variable models obey $|S| \leq 2$. Quantum mechanics allows $|S| \leq 2\sqrt{2}$, the Tsirelson bound. The substrate models are probed to see where in this range they fall.

B. Gauss sweep

In the *Gauss sweep*, the strength λ_G of the local Gauss-like constraint is varied over a set of values (e.g. $\lambda_G = 0, 0.5, \dots, 5$), while other parameters in the Hamiltonian are held fixed. For each λ_G , the ground state or a well-controlled low-energy state of the substrate model is prepared and the CHSH parameter $S(\lambda_G)$ is computed from the corresponding correlators.

The sweep probes whether Bell violations are sensitive to the enforcement of local consistency. If $|S|$ remains below 2 for all λ_G , the constrained substrate would behave classically with respect to CHSH. If there exists a region where $2 < |S| \leq 2\sqrt{2}$, the substrate supports nonclassical correlations. If $|S|$ were to exceed $2\sqrt{2}$, the model would be “superquantum” in the Popescu–Rohrlich sense.

C. Exchange sweep

In the *exchange sweep*, an exchange-like coupling J_{exch} between two effective spin sites is varied. At the substrate level, this is implemented by an additional Hamiltonian term that couples spin degrees of freedom in regions A and B . Again, for each value of J_{exch} we compute the CHSH parameter $S(J_{\text{exch}})$.

This sweep probes how Bell correlations depend on an explicit interaction channel between the two regions. A priori, one might expect strong correlations only when the interaction is large; the numerical results show a more nuanced picture.

D. Sequential perturbation test

The *sequential perturbation test* is designed to study how Bell correlations respond to a series of small disturbances at the substrate level. We begin from a configuration whose CHSH parameter is near the Tsirelson bound. At each step k :

1. A small random perturbation is added to the Hamiltonian or the state, injecting a controlled amount of energy ΔE_k .
2. The new state is evolved (if necessary) and its CHSH parameter S_k is computed.
3. The overlap with the original state is measured, $O_k = |\langle \psi_0 | \psi_k \rangle|$.

We then analyze the correlations between O_k , $|\Delta E_k|$, and $|S_k - S_0|$ across the sequence of steps.

E. Information extraction test

The *information extraction test* introduces an explicit classical readout channel coupled to the substrate. Operationally, we define a set of projectors or POVM elements $\{M_m\}$ on part of the Hilbert space. Each measurement outcome m yields classical information about the underlying configuration. In the simulations, the readout strength is dialed via a parameter that controls the degree of coupling between the substrate and an external register.

For a sequence of measurement steps k , we compute:

- the classical mutual information ΔI_k gained between the outcome stream and a chosen latent variable,
- the change in substrate entropy ΔS_k^{vN} ,
- the change in CHSH parameter ΔS_k .

We then compute aggregate quantities such as the total information gained, $\sum_k \Delta I_k$, the total entropy increase $\sum_k \Delta S_k^{\text{vN}}$, and correlations between ΔI_k and $|\Delta S_k|$.

F. Numerical implementation

All simulations are performed on finite Hilbert spaces of dimension small enough to permit exact diagonalization or Krylov-based solvers. The CHSH experiments use 2×2 lattices with a modest number of matter and link qubits; the three-quark bound-state example described later uses a $3 \times 3 \times 3$ lattice with a matrix-free Hamiltonian representation.

The scripts implement Hamiltonian construction, ground-state or low-energy state preparation, time evolution, and observable computation. Diagnostics summarize CHSH parameters, energy expectations, overlaps, and information-theoretic quantities, and produce visualizations of parameter sweeps.

IV. RESULTS: BELL CORRELATIONS AND CONSTRAINTS

A. Gauss sweep

In the Gauss sweep, the CHSH parameter $S(\lambda_G)$ exhibits three regimes as λ_G is increased from zero. At $\lambda_G = 0$ the Gauss penalty is inactive; the model nevertheless produces a clear Bell violation, with

$$|S(\lambda_G = 0)| > 2. \quad (6)$$

As λ_G is raised to moderate values, $|S|$ increases and approaches the Tsirelson bound. Numerically, the maximum observed value is

$$|S|_{\text{max}} \approx 2.8284, \quad (7)$$

achieved near $\lambda_G \approx 3$. Within numerical precision, this saturates the quantum Tsirelson limit $2\sqrt{2}$.

For very large λ_G , the Gauss constraint dominates the dynamics and the system is effectively frozen into low-violation subspaces. In this regime, $|S|$ decreases again and can fall back toward the classical bound. The overall picture is that of a broad plateau of λ_G values for which the constrained substrate supports robust Tsirelson-level correlations.

B. Exchange sweep

The exchange sweep explores how the CHSH parameter depends on an explicit exchange-like coupling J_{exch} between the two measurement regions. Remarkably, the model exhibits a Bell violation even when the exchange coupling is turned off:

$$|S(J_{\text{exch}} = 0)| \approx 2.8 > 2. \quad (8)$$

As J_{exch} is increased from zero, $|S|$ rises slightly and reaches a maximum of

$$|S|_{\text{max}} \approx 2.8284 \quad (9)$$

near $J_{\text{exch}} \approx 0.3$. Beyond this point, further increases in J_{exch} do not significantly enhance the Bell violation and can eventually disrupt the conditions that support Tsirelson-level behaviour.

The key observation is that the presence of a finite exchange channel is not strictly required to obtain strong Bell correlations in this substrate. Constraint structure alone is enough to generate nonclassical behaviour, and the exchange term fine-tunes the correlations within the quantum allowed region rather than moving the model into a qualitatively new regime.

C. Sequential perturbation test

In the sequential perturbation test, we start from a state with $|S| \approx 2.8$ and apply a series of small perturbations indexed by k . For each step, we record the entropy change ΔS_k^{vN} , the overlap O_k with the original state, and the change in CHSH parameter $\Delta S_k = S_k - S_0$.

Two empirical correlations emerge clearly:

- the overlap O_k is anti-correlated with the magnitude of the entropy increase $|\Delta S_k^{\text{vN}}|$,
- the overlap O_k is also anti-correlated with the magnitude of the change in CHSH parameter $|\Delta S_k|$.

In other words, larger perturbations tend to push the state further from its original configuration, both in Hilbert-space overlap and in terms of Bell correlations.

D. Simulation prediction: Bell deficit vs classical information

To make the information–disturbance tradeoff quantitative, we studied a minimal two-qubit model in which a Bell state is subjected to a sequence of weak σ_z measurements on one qubit. The initial state is $|\psi_+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, which saturates the Tsirelson bound $2\sqrt{2}$ for an appropriate choice of measurement settings. At each time step a tunable weak measurement with strength $\eta \in [0, 1]$ acts on qubit A , yielding a classical outcome and updating the state via a two-outcome Kraus channel. For each η , we run a fixed number of steps and compute: (i) the total classical mutual information $I_{\text{total}}(\eta)$ between the weak-measurement outcomes and an ideal σ_z outcome on qubit A , (ii) the final CHSH value $S(\eta)$, and (iii) the change in von Neumann entropy $\Delta S_{\text{vN}}(\eta)$ of the two-qubit state.

Figure 1 shows the resulting Tsirelson deficit $\Delta S(\eta) = 2\sqrt{2} - |S(\eta)|$ and entropy increase $\Delta S_{\text{vN}}(\eta)$ plotted as functions of $I_{\text{total}}(\eta)$ for a representative run with ten measurement steps and η values ranging from 0 to 0.5. Both quantities grow monotonically with the extracted information and are approximately linear in the small-information regime, before saturating as the state is driven toward a classical mixture. In particular, there is no regime in which substantial classical information is extracted while $|S|$ remains pinned at $2\sqrt{2}$.

In the small-information regime $I_{\text{total}} \lesssim 0.3$ bits, straight-line fits yield

$$\Delta S \approx a I_{\text{total}}, \quad \Delta S_{\text{vN}} \approx a' I_{\text{total}}, \quad (10)$$

with slopes a and a' of order unity in the natural simulation units. The precise numerical values depend on the details of the weak measurement channel and the number of steps, but the qualitative shape of Fig. 1—monotone, concave tradeoffs with an approximately linear onset—is robust across variants of the protocol.

outputs/info_tradeoff_sweep.png

FIG. 1. Tradeoff between Bell violation, entropy, and classical information in a toy two-qubit model. We start from the Bell state that saturates the Tsirelson bound and apply a sequence of weak σ_z measurements of tunable strength on one qubit. For each measurement strength we compute the total classical mutual information I_{total} extracted about an ideal σ_z outcome, the final CHSH value S , and the von Neumann entropy of the two-qubit state. **Left:** Tsirelson deficit $\Delta S = 2\sqrt{2} - |S|$ versus I_{total} . **Right:** entropy increase ΔS_{vN} versus I_{total} . Both quantities grow monotonically with the extracted information and are approximately linear in the small-information regime, before saturating as the measurement drives the state toward a classical mixture.

V. ADDITIONAL DEMONSTRATION: A THREE-QUARK BOUND STATE

To illustrate that the same substrate language can support bound-state-like structure, we construct a separate three-quark toy model on a finite spatial lattice. The model is not intended to reproduce any quantitative features of the proton; it serves as a testbed for how defrag-type potentials and constraints can stabilize localized composite objects.

A. Matrix-free three-quark Hamiltonian

The quark degrees of freedom are represented as three distinguishable spin- $\frac{1}{2}$ particles hopping on a cubic lattice of size $L_x \times L_y \times L_z$. Each quark can occupy any of the $N_s = L_x L_y L_z$ sites and has two spin states. The total Hilbert space dimension is therefore

$$\dim = (2N_s)^3. \quad (11)$$

For the simulations reported here, we use a $3 \times 3 \times 3$ lattice, so $N_s = 27$ and $\dim = 157,464$, which is tractable with matrix-free Krylov methods.

The Hamiltonian has the schematic form

$$H = H_{\text{hop}} + H_{\text{mass}} + H_{\text{defrag}} + H_{\text{Gauss}} + H_{\text{Zeeman}}. \quad (12)$$

Here H_{hop} describes nearest-neighbour hopping with strength J_{hop} ; H_{mass} assigns a mass m to each quark; H_{defrag} implements a spatially localized potential that favours central occupation; H_{Gauss} penalizes deviations from a target occupancy profile; and H_{Zeeman} couples the spins to an external field B . To keep the matrix dimension manageable, spin-exchange terms are omitted in this first pass.

Instead of constructing H explicitly as a dense or sparse matrix, we implement it as a matrix-free linear operator acting on the full state vector. The wavefunction is stored as a tensor $\psi_{r_1 s_1, r_2 s_2, r_3 s_3}$ with three position indices and three spin indices. Hamiltonian-vector products $H\psi$ are computed by applying each term in turn using vectorized operations, and the ground state is obtained via an iterative eigensolver.

B. Ground-state properties

For representative parameters

$$L_x = L_y = L_z = 3, \quad J_{\text{hop}} = 1.0, \quad m = 0.1, \quad g_{\text{defrag}} = 2.0, \quad \sigma_{\text{defrag}} = 0.7, \quad \lambda_G = 5.0, \quad B = 0, \quad (13)$$

the ground state has energy $E_0 \approx -11.92$ in the chosen units. The one-body density field $n(\mathbf{r})$ is obtained by summing $|\psi|^2$ over all but one of the position indices and all spins, and normalized so that $\sum_{\mathbf{r}} n(\mathbf{r}) = 3$.

On the $3 \times 3 \times 3$ lattice, the density profile is strongly peaked at the central site $\mathbf{r} = (1, 1, 1)$, with

$$n(1, 1, 1) \approx 0.23 \quad (14)$$

and smaller values at neighbouring sites. The profile is approximately spherically symmetric in lattice coordinates. A simple estimate of the effective radius,

$$r_{\text{eff}} = \sqrt{\frac{1}{3} \sum_{\mathbf{r}} n(\mathbf{r}) |\mathbf{r} - \mathbf{r}_{\text{cm}}|^2}, \quad (15)$$

where $\mathbf{r}_{\text{cm}}$ is the density barycenter, yields $r_{\text{eff}} \approx 1.77$ in lattice units. The barycenter itself lies at $\mathbf{r}_{\text{cm}} \approx (0.33, 0.33, 0.33)$ in the chosen coordinate system.

The total spin of the three quarks exhibits a nontrivial distribution. Defining S_z as the sum of the individual spin- $\frac{1}{2}$ components, the expectation values are

$$\langle S_z \rangle \approx 0.033, \quad \langle S_z^2 \rangle \approx 0.60, \quad (16)$$

with a histogram over the eigenvalues $S_z \in \{-3/2, -1/2, +1/2, +3/2\}$ that includes both doublet-like and quartet-like contributions. These numbers are not meant to match any real hadron; they simply show that spin structure and spatial localization coexist naturally in the constraint-based substrate.

C. Wavelet analysis (brief)

Because the proton density field lives on a regular lattice, it is natural to analyze it using discrete wavelet transforms. Preliminary three-dimensional wavelet decompositions of the density and spin fields indicate that most of the power resides in coarse scales, with smaller contributions from detail bands that encode the central peak and its immediate neighbours. A more thorough analysis of how these wavelet spectra evolve with parameters such as g_{defrag} and λ_G is left for future work, but the initial results confirm that the proton-like bound state is a genuine multi-scale structure in the substrate.

VI. EXPERIMENTAL PREDICTION

The toy models studied in this work suggest a concrete experimental prediction that goes beyond standard “yes/no” Bell tests and probes the cost of extracting additional classical information from a Tsirelson-level device.

In the information-extraction simulations, we coupled an explicit classical readout channel to part of the substrate and observed that, step by step, three quantities were positively correlated: (i) the classical mutual information ΔI_k gained about the underlying configuration, (ii) the entropy or heat ΔS_k^{vN} dumped into the substrate (and thence into whatever acts as its environment), and (iii) the reduction in the CHSH parameter $|S_k|$ away from its initial near-Tsirelson value. No free “peek” at the substrate was possible: whenever the readout learned something useful about the hidden configuration, the substrate paid both an entropic and a correlational price.

This motivates the following hypothesis, stated here as an experimentally falsifiable prediction:

Prediction. *Consider any physical implementation of a CHSH experiment that achieves a Bell parameter $|S|$ within a small tolerance of the Tsirelson bound $2\sqrt{2}$. Augment the setup by adding a controllable, local, weak measurement channel on one wing (for example, a continuous monitoring of one qubit or a tunable leakage port), and from its output construct a classical bit string that carries ΔI bits of mutual information per trial about some latent variable correlated with the pair of outcomes. Then, in the regime where the added channel does not itself dominate the dynamics, the following behaviour should be observed:*

1. *The average heat dissipated per trial into the monitored device plus environment, $\langle \Delta Q \rangle$, increases monotonically with the extracted information ΔI , with a lower bound of the Landauer form*

$$\langle \Delta Q \rangle \gtrsim k_B T_{\text{eff}} (\ln 2) \Delta I, \quad (17)$$

where T_{eff} is an appropriate effective temperature of the dissipative degrees of freedom.

2. *The Bell parameter $|S|$ decreases monotonically as a function of ΔI . To leading order in the small-information regime, the substrate framework predicts an approximately linear trade-off,*

$$2\sqrt{2} - |S(\Delta I)| \propto \Delta I, \quad (18)$$

with a proportionality constant set by microscopic details of the implementation but no regime in which $|S| \approx 2\sqrt{2}$ is maintained while ΔI is increased for free.

In other words, in a real CHSH experiment tuned close to the Tsirelson bound, any attempt to siphon off a small amount of classical side information about “what the system is doing”—via a weak probe, a tunable loss channel, or continuous monitoring—should be accompanied by a measurable increase in dissipation in the device and a correlated reduction in the observed Bell violation. The first part of the prediction is a Landauer-style lower bound on the energetic cost per bit of extracted information; the second part is a specific statement about how that cost shows up in the CHSH parameter.

VII. DISCUSSION

The numerical experiments presented here support a simple but consequential claim: *Tsirelson-level Bell correlations and bound-state-like patterns can emerge from explicitly finite, local quantum substrates equipped with constraint operators, without building the Born rule or additional nonlocal primitives into the microdynamics.*

In the Bell experiments, the Gauss and exchange sweeps show that violations of the CHSH inequality are not finely tuned artefacts. Across wide ranges of the constraint strength λ_G and the exchange coupling J_{exch} , the CHSH parameter $|S|$ takes values between the classical bound and the Tsirelson limit. There is a robust plateau where $|S| \approx 2\sqrt{2}$ within numerical precision. From the substrate perspective, this indicates that the combination of local Hamiltonian dynamics and Gauss-like consistency penalties is sufficient to carve out a region of state space with quantum-like correlations.

The sequential perturbation and information-extraction tests emphasize the cost of maintaining such correlations. Random perturbations that inject entropy into the substrate decrease the overlap with the original state and reduce $|S|$, with clear negative correlations between overlap and both $|\Delta S^{\text{vN}}|$ and $|\Delta S|$. Coupling an explicit readout channel shows the same trade-off in more operational terms: gaining classical information about the underlying configuration requires entropy production and degrades the Bell violation. No collapse postulate is invoked; the effect arises because the constraint structure that supports Tsirelson-level behaviour is disturbed when the substrate is forced to encode additional classical information.

These findings fit comfortably within the standard quantum-mechanical picture, where projective measurements and environmental decoherence also reduce Bell violations. The difference here is one of emphasis: instead of postulating an abstract measurement axiom, we model both the correlations and their fragility as consequences of concrete constraint-driven dynamics in a finite Hilbert space. From this vantage point, the Tsirelson bound and the usual information–disturbance trade-offs look less like arbitrary limits and more like features of a broader landscape of constrained substrates.

The two-qubit information sweep in Sec. IV D connects directly to the experimental prediction in Sec. VI. It shows that in a minimal model the deficit $2\sqrt{2} - |S|$ and the entropy increase ΔS_{vN} are both monotone, approximately linear functions of the extracted classical information in the small-information regime, and that there is no parameter region in which one can extract substantial information while keeping $|S|$ pinned at the Tsirelson bound. A laboratory test that mapped out the same curves for a high-fidelity photonic or superconducting-qubit Bell setup would therefore provide a direct check on the substrate picture.

The three-quark bound-state toy model points in a complementary direction. Using the same style of defrag potentials and Gauss-like penalties, we obtain a stable, centrally peaked density profile with nontrivial spin statistics on a $3 \times 3 \times 3$ lattice. While this model is far from realistic, it demonstrates that the framework can host both nonlocal Bell correlations and localized composite objects within a single unified language. In future work, it would be natural to explore whether the same mechanisms that stabilize the Bell plateau in parameter space also stabilize bound states against fragmentation and noise.

At the same time, the limitations of the present work are clear. The lattice sizes are small, the Hamiltonians are hand-crafted, and the models lack relativistic structure and gauge symmetry beyond simple $\mathbb{Z}_2$ ingredients. No attempt has been made to derive the toy constraints from deeper symmetry principles or to embed the models in a continuum field theory. The proton example is purely qualitative. Most importantly, nothing in these calculations demonstrates that the real world operates according to the substrate framework; at best they show that such a framework is *capable* of reproducing selected features of quantum behaviour.

Future work should therefore proceed along three main lines. First, it will be important to scale up the finite substrates, both in lattice size and in the richness of their local Hilbert spaces, using matrix-free and tensor-network techniques to keep the numerics tractable. Second, the range of phenomena studied should be broadened to include multi-party Bell tests, contextuality scenarios, and dynamical processes such as scattering and thermalization. Third, the interpretive side of the framework must be sharpened by identifying concrete experimental signatures that would distinguish constraint-based emergent quantum behaviour from more conventional unitary-plus-decoherence stories.

Despite these caveats, the results here provide a useful milestone. They show that a simple, finite, constraint-driven substrate can sit comfortably within the quantum correlation polytope, reach the Tsirelson bound, and exhibit realistic-looking trade-offs between correlation, entropy, and classical information flow. If nothing else, this gives the substrate program a firm numerical foothold from which to explore more ambitious connections to real-world physics.

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